

HexaBot – Autonomous Cleaning Bot

Department of Electronics & Telecommunication Engineering (SEE)

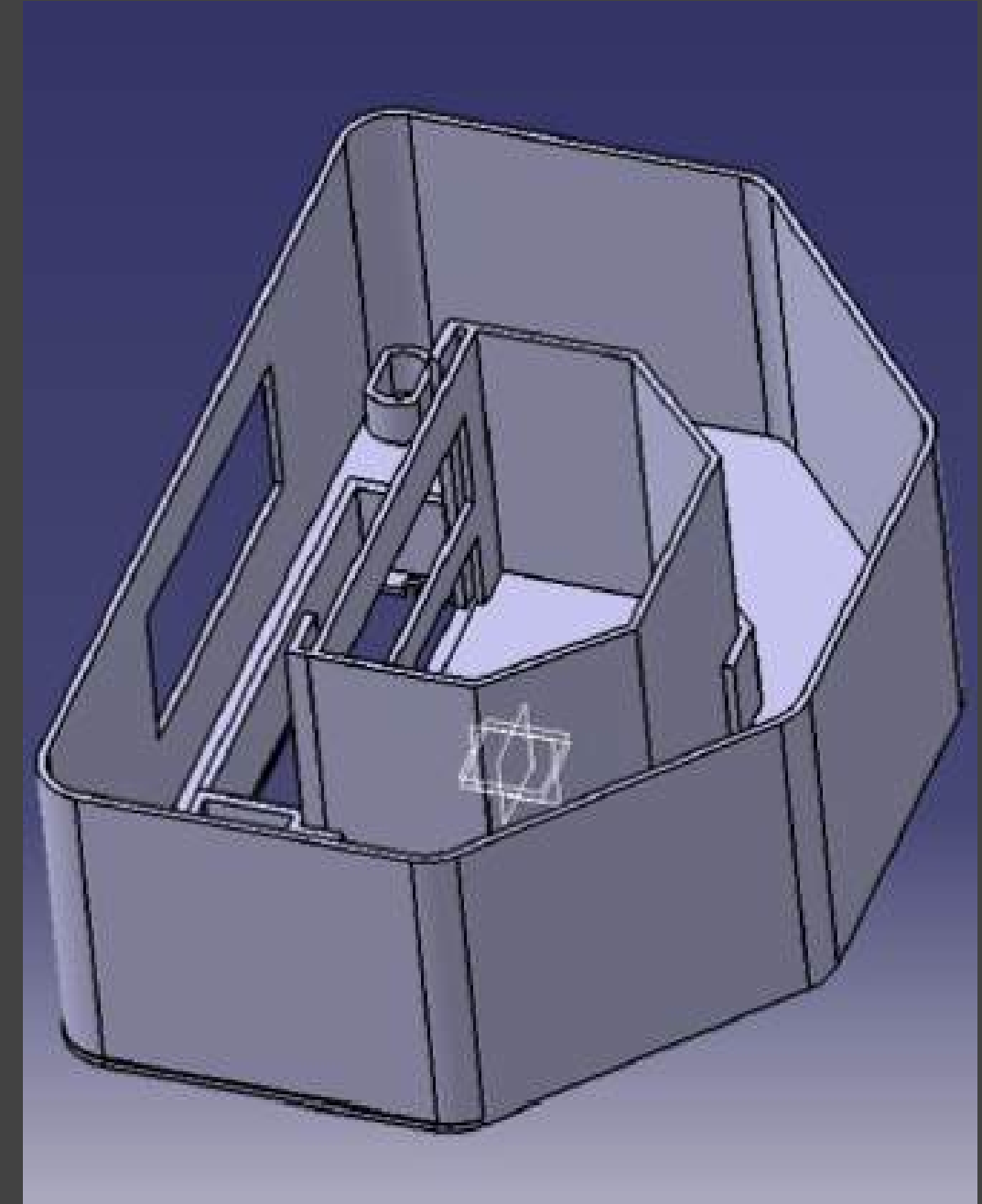
SY B.Tech Project Design

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Problem Statement & Motivation

- Manual cleaning processes are often time-consuming, labor-intensive, and inconsistent, especially in environments requiring high hygiene standards. The need for an autonomous, cost-effective, and efficient cleaning solution has increased with advancements in embedded systems. HexaBot is developed to address these challenges by providing a reliable robotic alternative capable of operating independently.



Scope of Project: Second Year (Sem IV)

01 System Integration

Integration of all hardware components including sensors, motors, and controller into a unified system.

Ensures smooth communication and coordinated operation between subsystems.

02 Navigation & Control Implementation

Development of obstacle avoidance and path control algorithms using sensor data.

Enables the robot to move autonomously with stable and efficient navigation.

03 Cleaning Mechanism Optimization

Improvement of brushing, suction, and mopping mechanisms for better performance.

Focuses on maximizing cleaning efficiency while maintaining compact design.

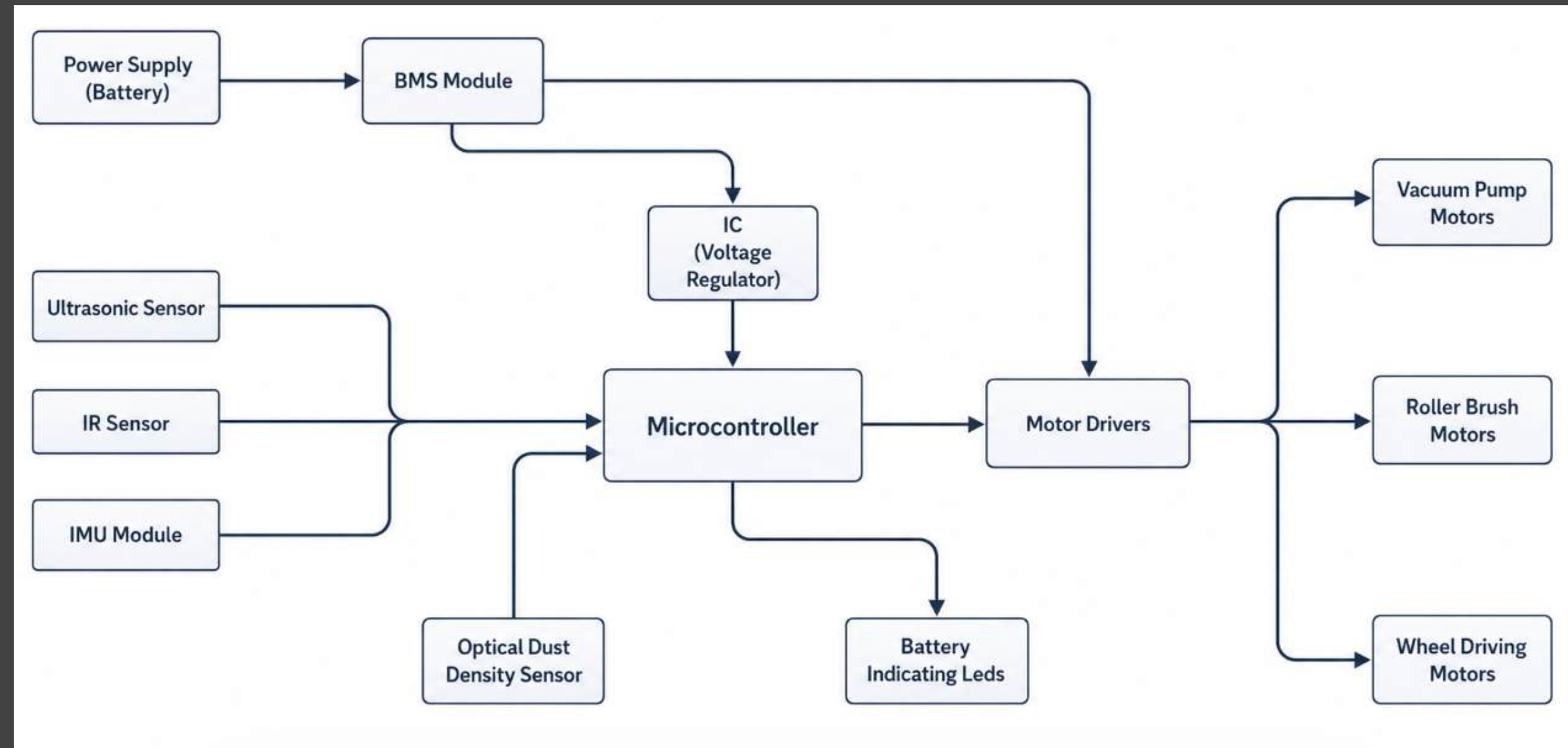
04 Testing & Performance Evaluation

Evaluation of system performance under different indoor conditions.

Includes efficiency analysis, runtime testing, and reliability assessment.

System Overview

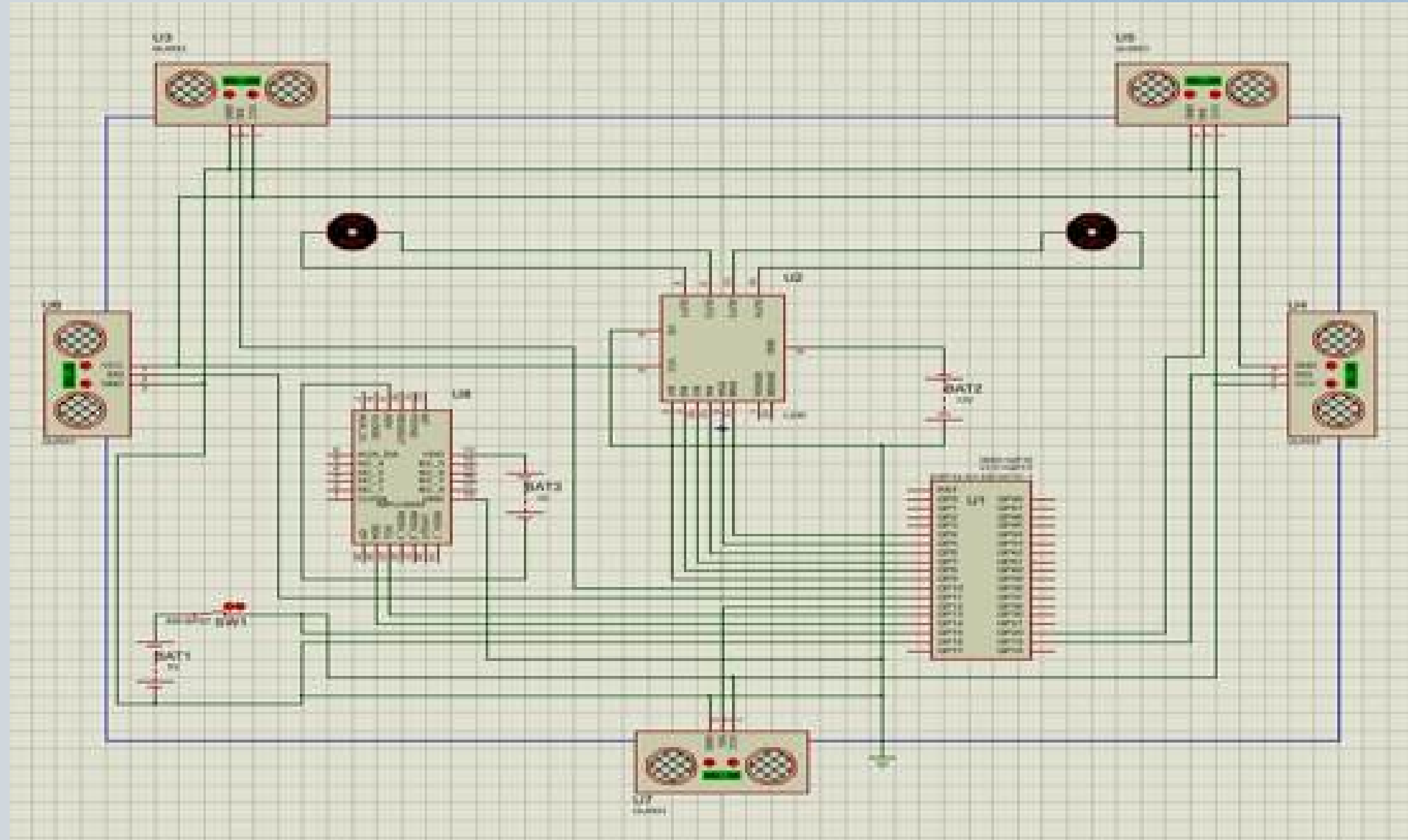
HexaBot is built around an ESP32 microcontroller that coordinates sensing, decision-making, and actuation. The robot employs a multi-stage cleaning approach that includes mechanical brushing, vacuum suction, and mopping to ensure thorough cleaning. A closed-loop control system continuously processes sensor data to adapt movement and cleaning actions in real time.



Circuit Diagram & Electrical Design

The circuit diagram shows the detailed electrical connections of the HexaBot system, including the ESP32 controller, motor drivers, sensors, and power supply. The system uses a lithium-ion battery with a BMS for safe operation, along with voltage regulators to provide stable 5V and 3.3V outputs.

Motor drivers and MOSFET circuits control the drive, brush, and suction motors, while sensors are interfaced with the controller for real-time input. Protection components such as capacitors and flyback diodes ensure reliable and stable performance.



Hardware Components

The hardware design consists of an ESP32 controller, motor drivers, multiple sensors, and high-efficiency DC motors for movement and cleaning. The system also includes a lithium-ion battery with a battery management system for safe operation. Each component is carefully selected to ensure optimal performance, reliability, and compact design.



Working Principle

HexaBot operates on a Sense–Think–Act control strategy. The robot continuously senses its environment using ultrasonic and infrared sensors, processes the data using embedded algorithms, and performs appropriate actions such as obstacle avoidance, directional movement, and activation of cleaning mechanisms. This real-time feedback system ensures efficient navigation and consistent cleaning performance.

Step No.	Process / Decision	Action
1	Start	Power ON the system
2	Initialize System	Initialize ESP32, sensors, and motors
3	System Check	Verify all components are working
4	Decision	If system error → Stop / Indicate error
5	Start Cleaning	Activate cleaning mechanisms
6	Read Sensors	Ultrasonic, IR, IMU, Dust sensor data
7	Decision: Edge Detected?	If yes → Stop, reverse, and turn
8	Decision: Obstacle Detected?	If yes → Avoid obstacle and change path
9	Decision: High Dust Detected?	If yes → Activate spot cleaning mode
10	Move & Clean	Continue forward movement with cleaning
11	Decision: Battery Low?	If yes → Stop operation
12	Loop	Repeat sensing and cleaning process
13	End	System stops

Conclusion & Future Scope •

HexaBot successfully demonstrates the integration of embedded systems, sensor-based navigation, and multi-stage cleaning into a compact and efficient robotic platform. The system achieves reliable performance and meets the intended design objectives. Future enhancements may include advanced navigation techniques, mobile application control, automatic charging systems, and IoT-based monitoring to further improve functionality and usability.

References

Books

1. Siegwart, R., Nourbakhsh, I., Scaramuzza, D.
2. Introduction to Autonomous Mobile Robots, MIT Press

Research Papers

2. Anish, N., Kumar, R. – Autonomous Floor Cleaning Robot Using Sensor Fusion Techniques (IJRR)
3. Tian, Y., et al. – Indoor Navigation Techniques for Service Robots (IEEE Access)

Technical Resources

4. ESP32 Docs – <https://www.espressif.com/en/products/socs/esp32>
5. MPU6050 – <https://invensense.tdk.com/products/motion-tracking/6-axis/mpu-6050/>
6. HC-SR04 – <https://lastminuteengineers.com/ultrasonic-sensor-arduino-tutorial/>
7. IR Sensor (TCRT5000) – <https://components101.com/>
8. 775 DC Motor – <https://www.electronicscomp.com/>
9. L298N Driver – <https://www.ti.com/lit/ds/symlink/l298.pdf>

Guides

10. Cleaning Mechanisms – <https://www.roboticsbusinessreview.com/>
11. Li-ion Battery Mgmt – <https://batteryuniversity.com/>